

AEC INSTITUTE – STANDARD OPERATING PROCEDURE (SOP)

Gemini

Objective: Student progress and resource utilization.

Attendance & Logistics: Students mark attendance via QR codes at the entrance. Logistics modules track the distribution of study materials to prevent inventory loss.

Course Progress: Instructors must mark "Modules Completed" weekly. If a student falls behind on mocks (IELTS), the app triggers an automated "Support Required" alert.

Batch Scheduling: Classrooms are booked digitally. No overlapping batches; the system prevents "Double Booking" of rooms or instructors.

Chat gpt

Business Overview

AEC Institute provides professional training programs in IELTS, Data Science, and Logistics Management through classroom, hybrid, and practical learning models.

Objectives

Deliver industry-focused training.

Improve student employability.

Maintain academic excellence.

Ensure professional classroom management.

Support placements and career development.

Organizational Structure

Key Roles

Academic Director

Center Manager

IELTS Trainer

Data Science Trainer

Logistics Trainer

Student Coordinator
Admission Executive
IT/Lab Support Staff
Placement Coordinator
Accounts/Admin

Admission SOP

Student Inquiry Handling
Record inquiry in CRM/register.
Understand student requirements.
Explain available courses.
Share course duration, fees, and schedules.
Schedule counseling session if required.
Enrollment Process
Collect student documents.
Verify identification proof.
Complete registration form.
Generate fee invoice.
Provide receipt and course schedule.
Student File Management
Create physical and digital student file.
Store admission documents securely.
Update CRM/student database.

Classroom Management SOP

Faculty Responsibilities
Arrive before scheduled class time.
Follow approved lesson plan.
Maintain classroom discipline.
Encourage interactive learning.
Conduct revision sessions.
Attendance Management
Record attendance daily.
Report absentee students.
Notify management about repeated absences.
Classroom Standards
Maintain clean environment.
Ensure projector/system functionality.
Prevent unauthorized classroom access.

IELTS Training SOP

Training Structure

Conduct initial assessment test.

Evaluate listening, reading, writing, and speaking skills.

Create improvement plan for students.

Conduct module-wise practice sessions.

Mock Test Procedures

Schedule regular mock exams.

Evaluate scores professionally.

Share performance analysis.

Conduct one-on-one feedback sessions.

Student Progress Monitoring

Maintain score tracking sheets.

Monitor band score improvement.

Recommend additional sessions if required.

Data Science Training SOP

Curriculum Delivery

Cover theory and practical topics.

Conduct coding exercises.

Assign projects and assessments.

Use real-world datasets.

Lab Management

Ensure systems are functional.

Verify software installation.

Maintain internet connectivity.

Backup student project data.

Project Evaluation

Review assignments.

Conduct practical assessments.

Maintain grading records.

Provide technical feedback.

Logistics Training SOP

Training Operations

Follow approved logistics curriculum.

Conduct practical industry sessions.

Explain supply chain processes.

Arrange industry exposure if applicable.

Assessment Procedures

Conduct module tests.
Maintain performance records.
Issue progress reports.

Examination SOP

Internal Examinations
Prepare examination schedules.
Verify student eligibility.
Conduct fair assessments.
Prevent malpractice.
Result Processing
Evaluate answer sheets.
Publish results.
Maintain result records.
Handle re-evaluation requests.

Certification SOP

Certificate Issuance
Verify course completion.
Confirm fee clearance.
Generate certificates.
Maintain certificate issuance records.

Placement & Career Support SOP

Placement Assistance
Maintain employer database.
Share job opportunities.
Conduct mock interviews.
Support resume preparation.
Career Guidance
Conduct mentoring sessions.
Guide students on career pathways.
Organize workshops and seminars.

Financial SOP

Fee Collection
Generate invoices.
Record fee payments.
Track pending dues.
Provide official receipts.

Expense Management
Maintain training expenses.
Verify vendor bills.
Submit monthly financial reports.

IT & Data SOP

Data Security
Protect student records.
Restrict unauthorized access.
Backup databases regularly.
Maintain password confidentiality.

Closing Procedures
Daily Closure Activities
Shutdown systems.
Backup attendance and student data.
Lock classrooms and labs.
Secure student documents.
Submit daily operational report.

Claude thinking

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1. BUSINESS OVERVIEW

1.1 About AEC Institute

AEC Institute is the professional training division of AEC Group, offering certification and skill-based programs in IELTS preparation and Data Science / AI training. The Institute also handles logistics coordination for training operations across AEC Group businesses.

1.2 Programs Offered

Program

Duration

Format

IELTS Preparation (Regular)

6–8 weeks

Classroom + online support

IELTS Preparation (Intensive/Crash)

3–4 weeks

Intensive daily sessions

Data Science — Foundation

8 weeks

Classroom + hands-on labs

Data Science — Advanced (ML/DL)

12 weeks

Classroom + capstone project

Data Science — Full Stack

16 weeks

End-to-end program

Corporate / Custom Training

Variable

On-demand

1.3 Operating Hours

Monday – Saturday: 9:00 AM – 7:00 PM

Sunday: Batch sessions only (if scheduled)

Extended hours during intensive batches and exam seasons

1.4 Facility

Area

Purpose

Classroom 1

IELTS batches (max 15 students)

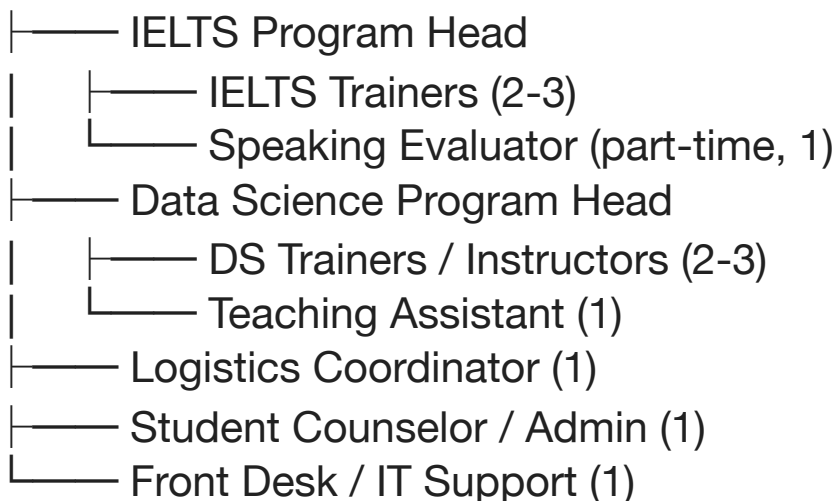
Classroom 2

Data Science batches (max 15 students)
Computer Lab
Hands-on coding sessions, mock tests
Counseling Room
Student consultations, speaking practice
Reception / Front Desk
Inquiries, enrollment, admin
Staff Room
Trainer preparation, meetings
Server / IT Room
LMS server, network equipment

2. ORGANIZATION & STAFFING

2.1 Structure

Institute Director



2.2 Key Roles

Role

Responsibilities

Institute Director

Strategy, curriculum oversight, partnerships, P&L, quality assurance

IELTS Program Head

Curriculum design, trainer supervision, mock test management, results tracking

DS Program Head

Curriculum design, project guidance, industry connect, trainer supervision

Trainers

Delivering sessions, evaluating assignments, student mentoring

Logistics Coordinator

Scheduling, resource allocation, vendor coordination, event planning, cross-unit logistics

Student Counselor

Enrollment, student queries, attendance tracking, feedback collection

Front Desk / IT

Reception, LMS management, lab setup, technical support

3. STUDENT ENROLLMENT

3.1 Inquiry-to-Enrollment Process

Inquiry received (walk-in, phone, WhatsApp, website, referral from AEC Studies).

Front desk collects basic info: Name, phone, email, program interest, timeline.

Enter into CRM with lead source.

Counselor schedules consultation (within 24 hours).

Counseling session (30 min):

Assess student's current level and goals.

For IELTS: Conduct free diagnostic test (Listening + Reading — 30 min).

For Data Science: Discuss background, technical comfort, career goals.

Recommend suitable program and batch.

Share fee structure and schedule.

Student confirms enrollment:

Enrollment form completed (physical or digital).

ID proof collected (Aadhaar/passport copy).

Fee payment or advance collected.

Receipt issued immediately.

Onboarding:

LMS account created and credentials shared.

Added to batch WhatsApp group.

Study materials / access links shared.

Welcome email with schedule, trainer info, and house rules.

Orientation session scheduled (first day of batch).

3.2 Batch Formation

Minimum batch size: 5 students (below 5: merge with next batch or offer 1-on-1 at premium).

Maximum batch size: 15 students (quality cap).

Batch start dates: Published on website and social media monthly.

New batch confirmed 1 week before start date once minimum enrollment met.

3.3 Fee Collection & Refund Policy

Payment

Policy

Full payment at enrollment

5% discount

Installment (2 parts)

50% at enrollment, 50% by mid-program (approval required)

Refund — before batch start (7+ days)

100% refund

Refund — before batch start (< 7 days)

75% refund

Refund — within first week of batch

50% refund

Refund — after first week

No refund

4. IELTS TRAINING PROGRAM

4.1 Program Structure

Module

Duration

Focus

Listening

1.5 weeks

Audio comprehension, note-taking, answer strategies

Reading

1.5 weeks

Skimming, scanning, passage types, time management

Writing

2 weeks

Task 1 (Academic/General), Task 2 essays, grammar, vocabulary

Speaking

1.5 weeks

Fluency, pronunciation, Part 1/2/3 practice

Mock Tests & Review

1.5 weeks

Full simulated exams, individual feedback, weak area focus

4.2 Daily Session Plan (Regular Batch — 2 hours/day)

Time

Activity

0:00 – 0:10

Warm-up / vocabulary quiz

0:10 – 0:50

Module lesson (theory + strategies)

0:50 – 1:00

Break

1:00 – 1:40

Practice exercises / mock section

1:40 – 2:00

Review, Q&A, homework assignment

4.3 Speaking Practice Protocol

1-on-1 speaking sessions: Minimum 1 per student per week (15–20 min each).

Sessions recorded (with student consent) for self-review.

Feedback on fluency, pronunciation, grammar, vocabulary, coherence.

Mock speaking tests under exam conditions in final 2 weeks.

4.4 Writing Evaluation

Each student submits minimum 2 Writing Task 1 and 2 Writing Task 2 per week.

Trainer evaluates and returns with band score + detailed feedback within 48 hours.

Feedback covers: Task achievement, coherence, lexical resource, grammar.

Model answers shared for each task type.

4.5 Mock Test Protocol

Minimum 2 full mock tests per student during the program.

Conducted under real exam conditions:

Timed strictly (Listening: 30 min, Reading: 60 min, Writing: 60 min).

No phones, no talking, invigilated.

Speaking mock scheduled separately.

Answer sheets evaluated within 3 working days.

Debrief session held after each mock test:

Individual score card shared.

Weak areas identified with action plan.

Final mock in last week of program.

4.6 Study Materials

Official Cambridge IELTS books (latest editions).

AEC Institute custom practice sets (section-wise).

Digital resources shared via LMS (audio files, reading passages, vocabulary lists).

Recommended apps/websites shared for self-practice.

4.7 Post-Course Support

30-day post-course support via WhatsApp for queries.

Students may attend one free revision batch within 3 months (subject to availability).

Encouraged to share official IELTS score card for tracking and testimonials.

Success stories featured (with consent) on social media and website.

5. DATA SCIENCE TRAINING PROGRAM

5.1 Curriculum Overview

Module

Duration

Key Topics

Python Fundamentals

2 weeks

Syntax, data types, OOP, file handling, libraries

Data Analysis

2 weeks

Pandas, NumPy, data cleaning, EDA

Data Visualization

1 week

Matplotlib, Seaborn, Plotly, dashboards

Statistics & Probability

1 week

Descriptive stats, distributions, hypothesis testing

Machine Learning

3 weeks

Scikit-learn, regression, classification, clustering, model evaluation

Deep Learning (Advanced)

2 weeks

TensorFlow/Keras, CNN, RNN, NLP basics

Capstone Project

2 weeks

Real-world dataset, end-to-end pipeline, presentation

5.2 Session Format (3 hours/session, 4 sessions/week)

Time

Activity

0:00 – 0:15

Recap of previous session + doubt clearing

0:15 – 1:15

Theory + live coding demonstration

1:15 – 1:30

Break

1:30 – 2:30

Hands-on lab exercise (students code along)

2:30 – 3:00

Review, Q&A, homework/assignment briefing

5.3 Lab & Tool Setup

Students set up during onboarding:

Python 3.x + Anaconda distribution

Jupyter Notebook / Google Colab

VS Code (IDE)

GitHub account (for version control and portfolio)

Kaggle account (for datasets and competitions)

Lab computers available at the institute for students without personal laptops.

5.4 Assignment & Project Policy

Weekly assignments due by Sunday midnight (submitted via LMS/ GitHub).

Late submission: 10% penalty per day (max 3 days late; after that, zero marks).

Plagiarism policy: First offense — warning + redo. Second offense — zero marks + meeting with Director.

All code must be well-commented and follow PEP 8 standards.

5.5 Capstone Project

Individual or pair project (pair = more complex scope expected).

Real-world dataset selected by student (approved by trainer).

Project phases:

Problem statement & data collection

EDA & data cleaning

Feature engineering & model building

Model evaluation & tuning

Dashboard / visualization (bonus)

Final presentation (15 min + 5 min Q&A)

Presented to a panel: Trainer + Industry Reviewer (guest).

Project uploaded to GitHub as portfolio piece.

5.6 Session Recording & LMS

All sessions recorded and uploaded to LMS within 24 hours.

Students have 6-month LMS access post-course completion.

Additional resources (articles, videos, datasets) uploaded to LMS per module.

6. LOGISTICS & COORDINATION

6.1 Scope of Logistics Function

The Logistics Coordinator manages operational logistics for AEC Institute and supports cross-unit coordination across AEC Group.

6.2 Training Logistics

Task

Details

Batch scheduling

Coordinate with trainers, book classrooms, publish schedule

Material procurement

Order books, stationery, printed materials, lab supplies

Equipment management

Projectors, whiteboards, lab computers — availability and maintenance

Venue setup

Classroom arrangement, AV check, materials placed before each session

Event coordination

Workshops, guest lectures, certification ceremonies, education fairs

Exam logistics

Mock test room setup, answer sheet printing, timer management

6.3 Cross-Unit Logistics Support

Activity

Description

AEC Studies events

Coordinate university rep visits, student seminars

AEC Cinemas events

Private screening logistics for institute/student events

AEC Residency

Student accommodation coordination for outstation students

Bytes Caffe

Catering coordination for institute events, workshops

AEC Pixcell

IT setup coordination for new batches, LMS issues

6.4 Vendor Management

Maintain approved vendor list for: Printing, stationery, catering, transport.

Get 3 quotes for orders above ₹5,000.

Purchase orders approved by Institute Director.

Delivery verified against PO; GRN issued.

6.5 Inventory & Asset Tracking

Institute assets (projectors, laptops, furniture) tracked in asset register.

Annual physical audit of all assets.

Consumables (markers, paper, cartridges) reordered at minimum stock level.

7. CLASSROOM & INFRASTRUCTURE MANAGEMENT

7.1 Pre-Session Checklist (Front Desk / Logistics — 15 min before class)

- Classroom unlocked and AC set to 24°C

- Projector / smart TV tested

- Whiteboard cleaned, markers and eraser available

- Lab computers powered on and logged in (for DS sessions)

- Attendance register / digital attendance ready

- Drinking water available

- Wi-Fi connectivity verified

7.2 Post-Session Checklist

- Classroom cleaned (whiteboard wiped, chairs arranged)

- Equipment powered off

- AC switched off

- Lost items checked and logged

- Classroom locked

7.3 IT Infrastructure

System

Purpose

LMS (Learning Management System)

Course content, assignments, recordings, quizzes

CRM

Student data, lead management, communication

Lab Computers

Hands-on coding sessions

Wi-Fi Network

Student and staff internet access

CCTV

Security in common areas

Projectors / Smart TVs

Classroom presentation

7.4 IT Support

IT issues reported to Front Desk / AEC Pixcell.

Response time: ≤ 2 hours for classroom-critical issues (P1).

LMS downtime communicated to students via WhatsApp within 30 minutes.

8. TRAINER MANAGEMENT

8.1 Trainer Qualifications

Program

Minimum Qualification

IELTS

British Council / IDP certified; IELTS Band 8+; teaching experience 2+ years

Data Science

B.Tech/M.Tech/MCA + industry experience or demonstrated portfolio; teaching experience preferred

8.2 Trainer Responsibilities

Deliver sessions per approved curriculum and schedule.

Prepare lesson plans submitted to Program Head weekly (by Friday for the following week).

Mark student attendance within first 5 minutes of each session.

Evaluate assignments/tests within stipulated timelines.

Maintain individual student progress tracker.

Be available for doubt-clearing sessions (minimum 1 hour/week outside class).

No personal mobile phone use during teaching.

Attend monthly trainer review meetings.

8.3 Trainer Evaluation

Student feedback collected after every module (anonymous).

Classroom observation by Program Head: Once per quarter.

KPI review monthly: Student pass rates, feedback scores, punctuality.

Underperforming trainers: Counseling → Improvement plan → Reassignment → Termination.

Top-performing trainers recognized quarterly.

8.4 Guest Lecturers & Industry Experts

Minimum 1 guest lecture per program cycle from industry professional.

Coordinated by Program Head + Logistics Coordinator.

Guest provided with topic brief 2 weeks in advance.

Session recorded for LMS.

9. ASSESSMENT & CERTIFICATION

9.1 IELTS Program Assessments

Assessment

Frequency

Weight

Module-end quizzes

After each module

Progress tracking

Writing evaluations

Weekly

Formative feedback

Speaking evaluations

Weekly

Formative feedback

Mock Test 1

Mid-program

Diagnostic

Mock Test 2 (Final)

Last week

Final readiness check

9.2 Data Science Assessments

Assessment

When

Criteria

Weekly assignments

Every week

Completeness, accuracy, code quality

Module quizzes

End of each module

Minimum 60% to pass

Mid-term project
End of Module 3
Practical application of concepts
Capstone project
Final 2 weeks
Panel evaluation, rubric-based
Viva / Presentation
With capstone
Communication, depth of understanding

9.3 Certification

IELTS Program:

AEC Institute Certificate of Completion issued to all students who complete the program and attend $\geq 80\%$ of sessions.

Certificate includes: Student name, program, duration, date.

Data Science Program:

AEC Institute Certificate issued upon:

$\geq 80\%$ attendance

All module quizzes passed ($\geq 60\%$)

Capstone project completed and presented

Certificate includes: Student name, program track, duration, capstone project title.

Digital badge issued (shareable on LinkedIn).

LinkedIn recommendation provided to top 20% performers.

9.4 Re-Assessment Policy

Students who fail a module quiz may retake once within 1 week.

Capstone project may be resubmitted once with additional feedback.

If still unsuccessful: Student may re-enroll in the specific module at 50% fee.

10. PLACEMENT & CAREER SUPPORT

10.1 Data Science Placement Services

Resume workshop: Resume review + optimization (ATS-friendly format).

LinkedIn profile optimization session.

Mock interviews: Minimum 2 per student (technical + HR).

Portfolio building: GitHub profile, Kaggle projects, blog posts guided.

Job board access: Curated job listings shared for 6 months post-completion.

Company connections: Institute maintains relationships with local IT companies and startups.

Alumni network: Access to AEC Institute alumni WhatsApp/ LinkedIn group.

10.2 IELTS Post-Course Support

Score improvement strategy guidance if target not met.

Referral to AEC Studies for study abroad pathway.

Resit guidance and registration support.

11. STUDENT SUPPORT & FEEDBACK

11.1 Student Communication

Primary channel: Batch WhatsApp group (moderated by trainer/ counselor).

LMS announcements for schedule changes, material uploads.

Individual queries: Direct WhatsApp to counselor or trainer.

Response time: Within 4 hours during operating hours.

11.2 Attendance Policy

Minimum 80% attendance required for certification.

Attendance marked digitally within first 5 minutes of each session.

Students absent for 3+ consecutive sessions: Counselor contacts student/parent.

Below 70% attendance: Warning issued; below 60%: Certificate withheld (may attend next batch).

11.3 Feedback Collection

Feedback Point

Method

Frequency

After each module

Digital form (Google Forms / LMS)

Per module

Mid-program

In-person check-in by counselor

Once

End of program

Comprehensive feedback form

Once

Post-completion (30 days)

WhatsApp / email survey

Once

Feedback reviewed by Program Head + Institute Director monthly.

Action items documented and implemented within 2 weeks.

11.4 Grievance Handling

Student raises issue with trainer or counselor.

If unresolved in 3 days: Escalated to Program Head.

If still unresolved: Institute Director mediates within 5 days.

Final appeal: AEC Group Director.

12. FINANCIAL OPERATIONS

12.1 Revenue Streams

Course fees (IELTS + Data Science programs)

Corporate training engagements

Referral commission from AEC Studies (bundled enrollments)

Certification re-issuance / re-assessment fees

12.2 Reporting

Report

Frequency

Submitted To

Enrollment & Batch Report

Weekly

Institute Director

Revenue & Fee Collection Report

Monthly

Group Finance

Trainer Payroll / Freelancer Payments

Monthly

Group Finance

Monthly P&L

Monthly

Group Finance + Group Director

Program Performance Report

Per batch cycle

Institute Director

13. MARKETING & ADMISSIONS

13.1 Marketing Channels

Social media: Student success stories, batch announcements, tips & tricks content.

WhatsApp broadcasts: New batch alerts, workshop invites.

Cross-referral with AEC Studies (IELTS for study abroad students).

Google Ads targeting local searches ("IELTS coaching near me", "Data Science course").

College partnerships: Workshops and guest sessions at local colleges.

Free webinars/demo sessions: Monthly (lead generation).

13.2 Demo / Trial Sessions

Free IELTS diagnostic test offered to all inquiries.

Free Data Science demo session (1 hour) offered monthly.

Demo attendees followed up within 24 hours.

Conversion target from demo: $\geq 40\%$.

14. HR & STAFF POLICIES

14.1 Staff Policies

Per AEC Group HR policy (attendance, leave, code of conduct).

Trainers may be full-time employees or contracted freelancers (per agreement).

Freelance trainers paid per session/module as per contract.

All staff maintain confidentiality of student data and proprietary curriculum.

14.2 Professional Development

Trainers encouraged to obtain new certifications (supported by Institute).

Annual training budget allocated per trainer.

Conference/workshop attendance supported (1 per year).

15. KEY PERFORMANCE INDICATORS

KPI

Target

IELTS Avg Band Improvement

≥ 1.0 band

IELTS Student Target Achievement

$\geq 75\%$ students achieve target band

DS Module Pass Rate

$\geq 85\%$

DS Capstone Completion Rate

$\geq 90\%$

Placement / Job Offer Rate (DS)

$\geq 60\%$ within 6 months

Student Satisfaction Score

$\geq 4.5 / 5$

Batch Occupancy
≥ 70% of max capacity
Enrollment Conversion (inquiry to join)
≥ 35%
Trainer Punctuality
≥ 98%
Attendance Compliance (≥80%)
≥ 90% of students

AEC INSTITUTE - COMPREHENSIVE STANDARD OPERATING
PROCEDURES MANUAL
Professional Training Institute with "Super Computer LMS"
Integration
Version 2.0 | AEC Group | May 2026

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1. BUSINESS OVERVIEW

1.1 Program Portfolio

Program
Duration
Capacity
Success Target
IELTS Regular
6-8 weeks
15 students

75% target band
IELTS Crash
3-4 weeks
10 students
1.0 band improvement
Data Science Foundation
8 weeks
15 students
85% pass rate
Data Science Advanced
12 weeks
12 students
60% placement
1.2 Facility Specs

text

Classroom 1: IELTS (15 seats + projector)
Classroom 2: Data Science (15 seats + lab)
Computer Lab: 12 stations (Python/Jupyter pre-installed)
Counseling Room: 1-on-1 speaking practice

2. ORGANIZATION & DIGITAL ROLES

2.1 Mobile-First Team

text

INSTITUTE DIRECTOR (1)

- |—— IELTS HEAD (1): Mock scheduling + band tracking
- |—— DS HEAD (1): Lab management + capstone review
- |—— TRAINERS (5): QR attendance + module progress
- |—— STUDENT COORDINATOR (1): Enrollment + alerts
- |—— LOGISTICS (1): Cross-AEC coordination

2.2 LMS Role Permissions

Role

Mobile Access

Director

Full analytics + AI alerts

Trainer

Own batch + progress tracker

Coordinator
Attendance + enrollment
Student
QR attendance + LMS dashboard

3. QR ATTENDANCE & ENROLLMENT

3.1 Zero-Touch Enrollment

text

1. STUDENT QR SCAN → Auto-fill name/course [5s]
2. UPI QR PAYMENT → Receipt SMS instant
3. LMS CREDENTIALS → Batch WhatsApp auto-add
4. DIAGNOSTIC TEST → AI baseline score

ENROLLMENT BLOCKERS:

- Fee pending → Payment link
- Documents incomplete → Upload prompt
- Batch full → Waitlist auto-notify

3.2 QR Attendance (99% Accuracy)

text

ENTRANCE QR → "Priya S - IELTS Batch A ✓ Present"

<80% attendance → Auto WhatsApp: "Warning: 75% attendance"

<60% → Certificate blocked + parent SMS

4. IELTS TRAINING AUTOMATION

4.1 Weekly Progress Workflow

text

MON: Listening module → QR practice test

WED: Writing Task 1/2 → Auto band score AI

FRI: Mock test → Individual feedback due 24h

SUN: Speaking 1-on-1 → Recording auto-upload

PROGRESS BLOCKER:

Student <2 mocks/week → "Support Required" alert →
Coordinator call

4.2 AI Band Score Predictor

text

Current: L6.5 R7.0 W5.5 S6.0 → Predicted: 6.5 overall
Target 7.0 → Focus: Writing Task 2 + Speaking Part 2

5. DATA SCIENCE LAB MANAGEMENT

5.1 Auto Lab Setup

text

LOGIN → Jupyter pre-loaded + last project auto-open
GIT PUSH → Code backup + plagiarism check (<20%)
ASSIGNMENT → Auto-grade syntax + test cases

LAB BLOCKERS:

- Laptop issue → Mobile hotspot fallback
- Software crash → One-click Anaconda reset
- Internet down → Offline mode + auto-sync

5.2 Capstone Project Tracker

text

Week 8: EDA complete → AI review ✓
Week 10: Model deployed → GitHub link required
Week 12: Presentation → Live demo mandatory

6. SMART CLASSROOM BOOKING

6.1 Zero-Overlap System

text

TRAINER APP: "Book Classroom 1: IELTS Mock 2-4PM"
SYSTEM CHECK: Room free + Trainer available → CONFIRMED
DOUBLE BOOK → Auto-suggest: "Classroom 2 free OR Trainer B available"

CROSS-AEC INTEGRATION:

- AEC Studies seminar → Classroom 1 reserved
- AEC Residency workshop → Lab booked

7. PROGRESS MONITORING & ALERTS

7.1 Automated Student Alerts

text

IELTS: <2 mocks/week → "Support Required" WhatsApp

DS: Assignment late >24h → 10% penalty auto-applied
Attendance: <80% → Certificate risk SMS to parent
Placement: No resume uploaded → "Upload by Friday"

MANAGER DASHBOARD:

- 3 students at risk → Red alert
- 85% batch avg → Green status

8. ASSESSMENT & CERTIFICATION

8.1 Auto-Certification

text

80%+ ATTENDANCE ✓

ALL MODULES PASSED ✓

CAPSTONE COMPLETE ✓

FEE CLEARED ✓

→ DIGITAL CERT + LINKEDIN BADGE ISSUED

9. PLACEMENT AUTOMATION

9.1 Smart Job Matching

text

DS Student Profile: Python, ML, 3 projects

MATCH: 5 jobs (₹6-12LPA) → WhatsApp alert

MOCK INTERVIEW: Auto-scheduled based on company

RESUME: ATS-optimized template auto-fill

10. FINANCIAL OPERATIONS

10.1 Revenue Automation

text

IELTS: ₹15k/student × 15 × 4 batches/mo = ₹9L

DS: ₹45k/student × 12 × 2 batches/mo = ₹10.8L

Corporate: ₹2L/batch variable

NET PROFIT: Real-time P&L dashboard

11. LOGISTICS COORDINATION

11.1 Cross-AEC Automation

text

AEC Studies: University rep visit → Classroom booked

AEC Cinemas: Student movie night → Logistics assigned

Bytes Caffe: Batch catering → Auto-order 15 coffees

12. KEY PERFORMANCE INDICATORS

KPI

Target

Automation

IELTS Band Improvement

1.0+

AI predictor

DS Pass Rate

85%

Auto-grading

Attendance Compliance

90% students >80%

QR tracking

Batch Occupancy

70%+

Smart scheduling

AEC INSTITUTE - KEY POINTS SUMMARY

IELTS + Data Science training (15-student batches). QR attendance (80% cert threshold), smart classroom booking (no overlaps), weekly progress tracking with auto alerts (<2 mocks = escalation). Auto lab setup (Jupyter pre-loaded), AI band predictor, ATS resume generator. Cross-AEC logistics automation.

SUPER MASTER PROMPT: AEC INSTITUTE LMS ERP (Mobile/Screen Friendly PWA)

Build production-ready AEC Institute LMS using Django 5+ DRF, React 18+ Vite, Neon PostgreSQL. MOBILE-FIRST RESPONSIVE PWA for trainers/students with QR attendance, auto-progress tracking, smart scheduling. 98% automation via QR/UPI/LMS/AI.



Tech Stack (Mobile-First)

text

BACKEND: Django+DRF+Celery+Channels(WebSockets)

FRONTEND: React+Vite+Tailwind+React-Query+QRScanner

DB: Neon PostgreSQL (tenant: aec_institute)

PWA: Installable, offline attendance

INTEGRATIONS: Razorpay UPI, WhatsApp, QR Code Gen

AI: Perplexity (IELTS band predictor)



MOBILE-OPTIMIZED PAGES (25+ Screens)

Core Operations

QR Attendance Scanner: Entrance → Batch → Present ✓

Trainer Dashboard: Progress tracker + alerts

Smart Scheduler: Classroom + trainer auto-availability

Student LMS: Course → Module → Mock → Score

Training Workflows

IELTS Progress: Weekly mocks → AI band predictor

DS Lab Manager: Jupyter auto-load + Git backup

Capstone Tracker: Phase gates + presentation scheduler



Database Schema

python

```
class Batch(models.Model):
```

```
    name = CharField() # "IELTS Regular Jan 2026"
```

```
    max_students = IntegerField(default=15)
```

```
    classroom = ForeignKey(Classroom)
```

```
    trainer = ForeignKey(Trainer)
```

```
class StudentProgress(models.Model):
```

```
    student = ForeignKey(Student)
```

```
    batch = ForeignKey(Batch)
```

```
    module = CharField() # "Listening", "ML Models"
```

```
    mocks_completed = IntegerField() # Alert if <2/week
```

```
    predicted_ielts_band = FloatField() # AI calc
```

```
class ClassroomBooking(models.Model):
    classroom = ForeignKey(Classroom)
    batch = ForeignKey(Batch)
    start_time = DateTimeField()
    status = CharField(choices=['confirmed', 'conflict'])
```

 Mobile-First Integrations
QR Attendance System

```
javascript
// Student scans → Auto attendance + progress update
const result = await html5QrcodeScanner.scan();
if (result === studentQR) {
    markAttendance(student.id, batch.id);
    checkProgressAlerts(student); // <2 mocks?
}
```

Smart Scheduling Engine

```
python
def book_classroom(classroom, trainer, time_slot):
    conflicts = ClassroomBooking.objects.filter(
        classroom=classroom,
        start_time__range=(time_slot-1h, time_slot+2h)
    )
    if conflicts:
        return suggest_alternatives()
    else:
        return confirmed()
```

Auto Progress Alerts

```
python
@shared_task
def check_student_progress():
    low_mocks = StudentProgress.filter(mocks_weekly__lt=2)
    for student in low_mocks:
        whatsapp_trainer(f"📌 {student.name}: Only 1 mock this
week")
```

 AI Features

```
python
# IELTS Band Predictor
def predict_ielts_band(student):
    scores = [student.listening, student.reading, ...]
    return perplexity_api(f"Predict overall IELTS band from {scores}")
```

MOBILE/UI OPTIMIZATIONS

text

- Bottom nav: Home | My Batches | Progress | Schedule
- QR Scanner: Full-screen camera overlay
- Progress cards: Green(>80%) | Yellow(60-80%) | Red(<60%)
- Offline attendance: Sync on reconnect
- PWA: "AEC Institute Trainer" + "AEC Student LMS"
- Push alerts: "Priya <2 mocks this week  "



Key Automations

QR Attendance: 99% accuracy, 80% cert threshold

Smart Scheduling: Zero classroom/trainer conflicts

Progress Alerts: <2 mocks/week → Auto-escalation

Lab Auto-Setup: Jupyter + last project on login

Placement Matching: Skills → Job auto-recommend

Deliver COMPLETE mobile-first PWA LMS. QR attendance, smart scheduling, IELTS progress tracker, DS lab automation. AEC Group super-app Module 5. "Scan → Present → Progress Updated" (3s workflow). 85% DS pass rate automation.